

SUPPLEMENTARY MATERIAL

Joint Scheduling of UAV Cargo Delivery and Communication Relay for Mountain Flood Disaster Response: A Case Study

Detailed sensitivity and robustness analysis

Repository: <https://github.com/Enyi-Gao/Mountain-Flood-UAV-Scheduling>

This supplement reports detailed scenario-level sensitivity tests that are summarized only briefly in the conference paper.

S1. Sensitivity-analysis protocol

The sensitivity study uses a one-factor-at-a-time design around the 80-box, two-relay-UAV, original-terrain reference scenario. To ensure comparable runs, the sensitivity mode disables the baseline-specific preset batching and release-time fallback and uses the same dynamic scheduling procedure across perturbed scenarios.

Joint feasibility criterion. A joint transport-relay solution is classified as feasible only when both (i) the number of hard delivery-deadline violations is zero and (ii) the number of communication-outage samples in the final 5 s audit is zero.

Demand perturbation. The number of relief boxes is varied over 60, 80, 100, and 120. The original demand list is resampled so that the spatial distribution, commodity types, priorities, and deadline attributes are approximately preserved. The relay fleet is fixed at two UAVs and the original terrain is retained.

Relay-fleet perturbation. The demand is fixed at 80 boxes and terrain is unchanged, while the number of available relay UAVs is varied from one to four.

Terrain perturbation. Terrain relief is scaled about the dispatch-center elevation according to $z' = z_{_O01} + \beta(z - z_{_O01})$, with $\beta = 0.8, 1.0$, and 1.2 . Service-area elevations are transformed consistently, while horizontal geometry, demand, and fleet resources are unchanged.

Interpretation of infeasible cases. When hard-deadline violations or communication outages remain, the reported makespan is retained only as a diagnostic output and is not interpreted as an improved feasible operating point.

S2. Relief-demand sensitivity

The demand-scale experiment reveals a clear transition from a robust operating region to overload. The 60- and 80-box cases remain jointly feasible. At 100 boxes, the transport schedule itself still satisfies all hard deadlines, but substantial soft lateness appears; after transport-relay coordination, six hard-deadline violations and 252 outage samples remain. At 120 boxes, hard-deadline violations already appear in the transport stage and the joint schedule deteriorates further.

Boxes	Sorties	Q2 hard	Q2 soft	Q2 makespan (s)	Q2 energy (kWh)
60	22	0	0	6646.4	60.35
80	26	0	0	7467.3	74.60
100	40	0	190670	12879.0	106.06
120	47	3	584940	14535.0	123.96

Table S1. Transport-stage response to relief-demand scaling.

Boxes	Outage samples	Relay sorties	Final hard	Final soft	Joint makespan (s)	Feasible
60	0	4	0	0.0	6871.2	Yes
80	0	4	0	1940.8	8281.2	Yes
100	252	9	6	1620100.0	37816.0	No
120	341	11	11	1565800.0	40684.0	No

Table S2. Joint transport-relay response to relief-demand scaling.

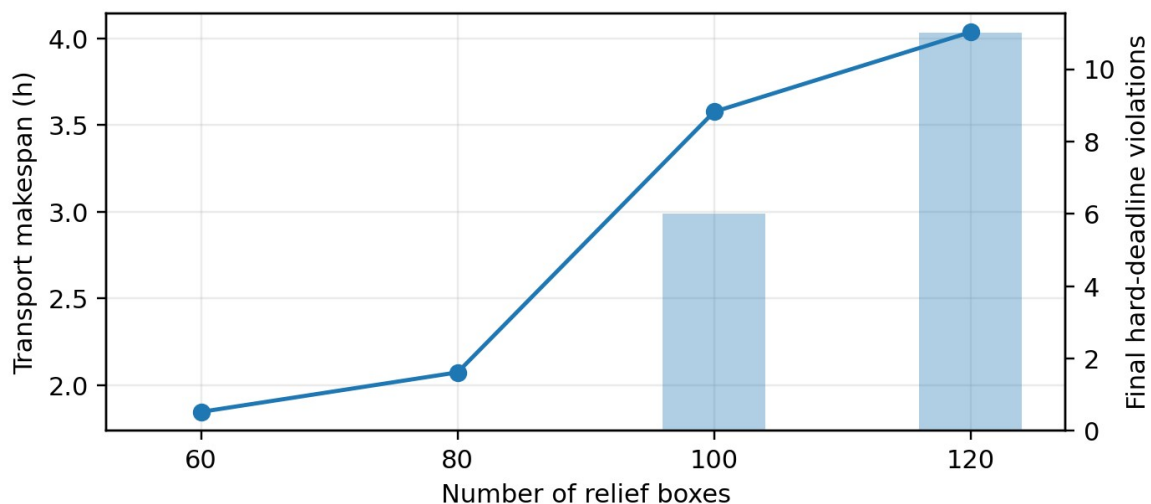


Figure S1. Demand sensitivity: transport makespan and final hard-deadline violations.

The main transition occurs between 80 and 100 boxes: hard transport deadlines remain satisfiable at 100 boxes, but the coupled communication-support problem becomes infeasible. At 120 boxes, the transport subsystem itself also crosses the hard-deadline boundary. This indicates that transport-communication coupling can become limiting before pure transport capacity is exhausted.

S3. Relay-UAV fleet sensitivity

Relay availability strongly affects whether communication support can be provided without damaging the delivery timeline. With one relay UAV, communication outages can be removed only through large transport delays, producing 11 hard-deadline violations. Two relay UAVs restore hard-constraint feasibility. Three relay UAVs eliminate the remaining soft lateness and reduce joint makespan, while a fourth relay UAV yields no further benefit.

Relay UAVs	Relay sorties	Outage samples	Final hard	Final soft	Joint makespan (s)	Relay energy (kWh)	Feasible
1	3	0	11	2051600.0	17194.0	5.05	No
2	4	0	0	1940.8	8281.2	4.49	Yes
3	4	0	0	0.0	7732.7	4.61	Yes
4	4	0	0	0.0	7732.7	4.61	Yes

Table S3. Sensitivity to the number of available relay UAVs.

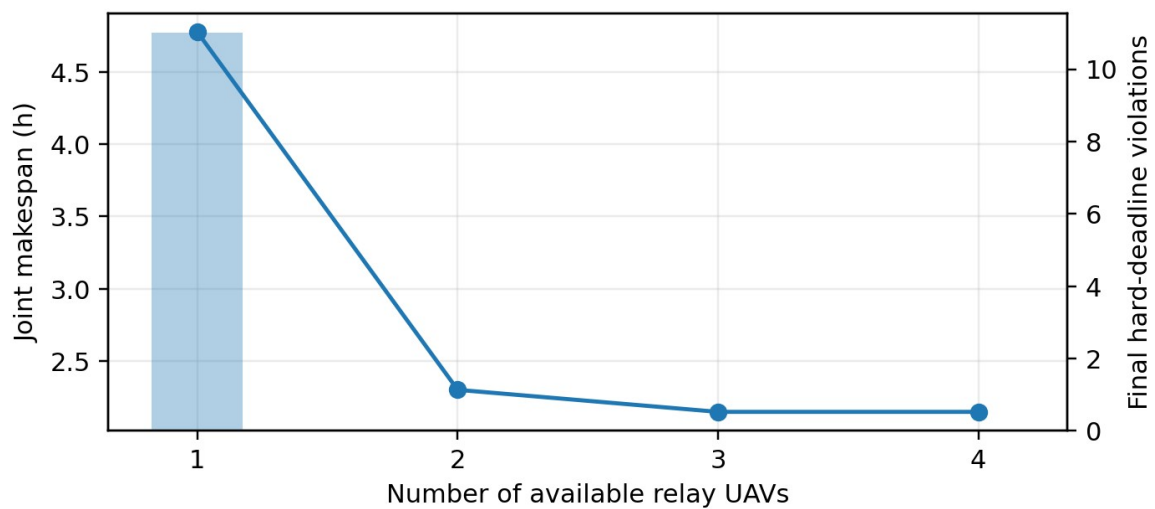


Figure S2. Relay-fleet sensitivity: joint makespan and final hard-deadline violations.

The results show a resource threshold and then diminishing returns. Two relay UAVs are sufficient to recover hard-constraint feasibility, three remove the remaining soft lateness, and the fourth airframe is redundant for this scenario because the required relay sorties can already be covered by the available concurrency.

S4. Terrain-relief sensitivity

Terrain relief is scaled around the dispatch-center elevation so that $\beta < 1$ compresses vertical relief and $\beta > 1$ amplifies it. Transport time and energy rise gradually from $\beta = 0.8$ to 1.2. The joint schedule remains feasible for $\beta = 0.8$ and 1.0. At $\beta = 1.2$, the transport-only schedule still has zero hard-deadline violations, but relay-related temporal repair creates four hard-deadline violations; the communication audit itself still reports zero uncovered samples.

beta	Q2 makespan (s)	Q2 energy (kWh)	Relay sorties	Outage samples	Final hard	Final soft	Joint makespan (s)	Feasible
0.8	7331.9	74.01	4	0	0	935.3	8155.2	Yes
1.0	7467.3	74.60	4	0	0	1940.8	8281.2	Yes
1.2	7602.7	75.20	5	0	4	10022.0	8051.8	No

Table S4. Sensitivity to terrain-relief scaling.

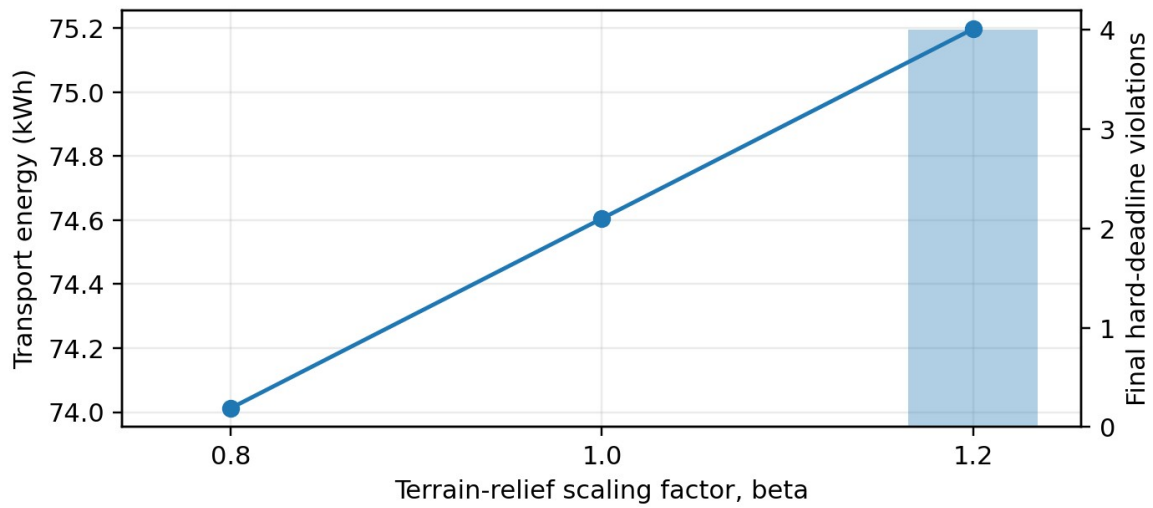


Figure S3. Terrain-relief sensitivity: transport energy and final hard-deadline violations.

Although the reported joint makespan at $\beta = 1.2$ is not larger than that at $\beta = 1.0$, the $\beta = 1.2$ solution is infeasible because four hard delivery deadlines are violated. Feasibility, rather than makespan alone, should therefore be treated as the primary robustness criterion near this boundary.

S5. Summary, practical uncertainties, and reproducibility

Across the three scenario-level sensitivity dimensions, the framework is robust under moderate perturbations but degrades nonlinearly near resource-capacity boundaries. Demand growth and insufficient relay concurrency produce the strongest deterioration. Terrain relief has a smaller effect in the moderate range but can also make the coupled schedule infeasible once relay-related timing adjustments begin to conflict with delivery deadlines.

The case study remains deterministic. Real disaster operations may additionally involve uncertain or time-varying demand arrivals, temporary UAV failures, wind-dependent flight time and energy consumption, battery aging, communication-channel fluctuations, weather-dependent visibility, and evolving landing or road conditions. These uncertainties motivate stochastic or robust optimization and online replanning in future work.

Reproducibility. The accompanying repository stores the sensitivity figures, numerical summaries, and MATLAB scripts used to generate the supplementary results. Repository: <https://github.com/Enyi-Gao/Mountain-Flood-UAV-Scheduling>